

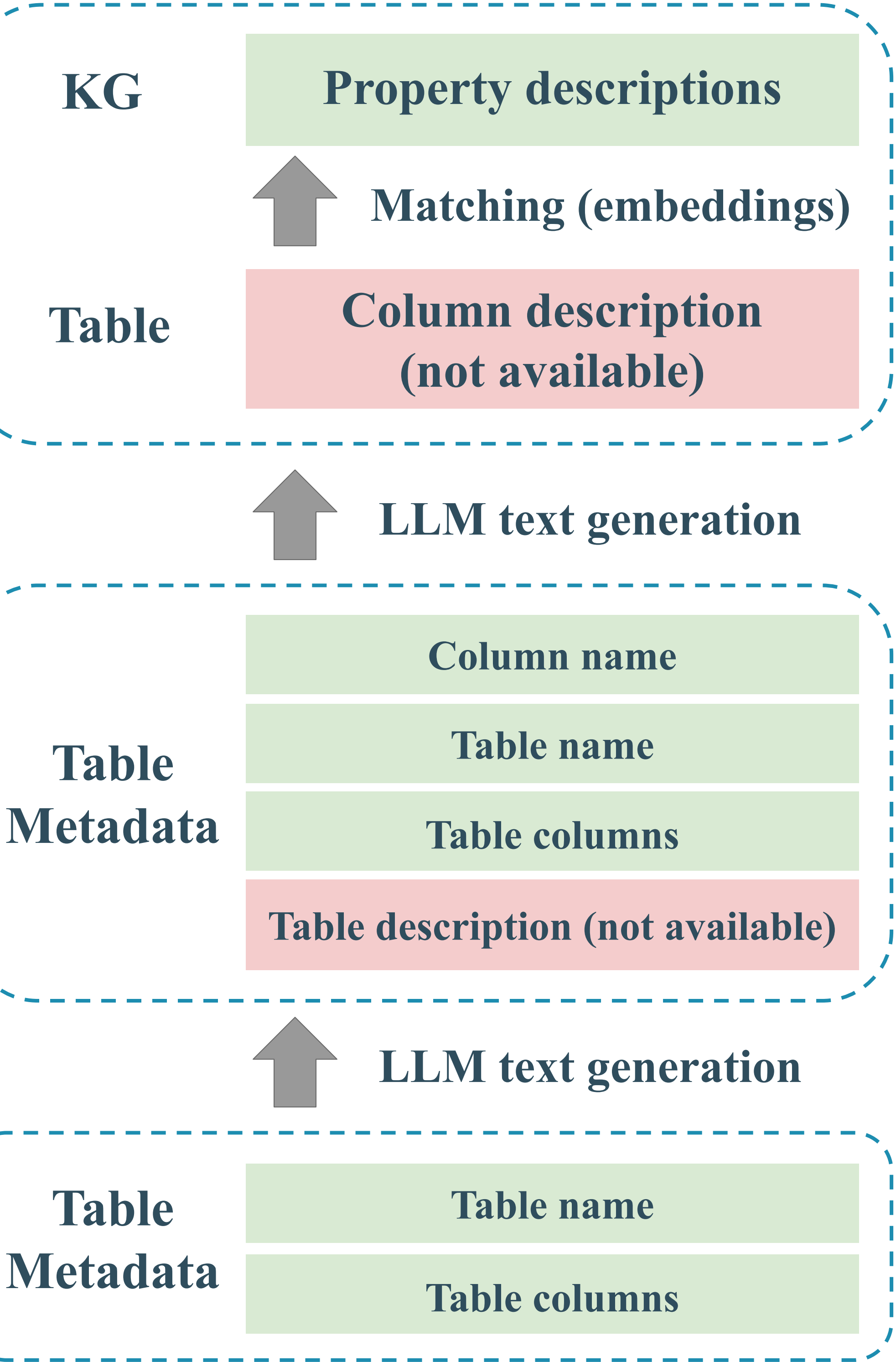
# Matching Table Metadata to Knowledge Graphs: A Data Augmentation Perspective

## Background & Research question

|                                |        |        |       |               |                                                                                                                                                                                                                                                          |
|--------------------------------|--------|--------|-------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Table name                     | Museum |        |       |               | Property - Description (from KG) <ul style="list-style-type: none"><li>"location" - "The location of the thing."</li><li>"locationName" - "Location of the thing as string."</li><li>"locationCity" - "City the thing is located."</li><li>...</li></ul> |
| Table columns                  | RANG   | Museum | Stadt | Facebook-Fans |                                                                                                                                                                                                                                                          |
| Table cells<br>(not available) |        |        |       |               |                                                                                                                                                                                                                                                          |

- Task & Background:** Given limited table metadata, without accessing the table's data or content, the task is to align each table column with a corresponding property from a knowledge graph. This alignment relies on embedding-based matching using text similarity, which requires well-defined descriptions on both table and knowledge graph.
- Research Question:** How can we generate high-quality description for each column considering the table context before performing embedding-based matching?

## Methodology



## Prompt & Example

Given the table name and corresponding table columns, your task is to look at all the following information, think about it step by step from the frequently-used terms in this topic, and generate a table description with one simple example for explanation in 3-4 sentences.

Table name: **Museum**

Table columns: ["RANG", "Museum", "Stadt", "Facebook-Fans"]

Table description:

The "Museum" table stores information about various museums across different cities. Each row represents a museum, with its ranking ("RANG"), name ("Museum"), location ("Stadt"), and number of Facebook fans ("Facebook-Fans"). This data allows for easy comparison and analysis of the popularity of museums based on their social media presence. For example, a top-ranked art museum like the Louvre in Paris might have a high "RANG" value, a large number of "Facebook-Fans"...

Given the table name, header, description, and one specified column name, your task is to think about it step by step from the frequently-used terms in this topic and generate a definition for this column in one short sentence with 10-15 words, please also make it easy to distinguish from the other columns.

Table name: **Museum**

Table columns: ["RANG", "Museum", "Stadt", "Facebook-Fans"]

Table description: The "Museum" table stores information about various museums across different cities. Each row represents a museum...

Column name "**Stadt**" definition:

The "Stadt" column represents the city where each museum is located.

## Preliminary Results & Findings

| Methods                              | Re-ranking | Dataset 1 |       | Dataset 2 |       |
|--------------------------------------|------------|-----------|-------|-----------|-------|
|                                      |            | Hit@1     | Hit@5 | Hit@1     | Hit@5 |
| SOTA<br>(Vandemoortele et al., 2024) | no         | 33%       | 52%   | 41%       | 72%   |
|                                      | yes        | 75%       | 92%   | 83%       | 98%   |
| Ours                                 | no         | 47%       | 63%   | 71%       | 94%   |

### Performance of our approach on the Metadata2KG datasets

The state-of-the-art (SOTA) method generates column descriptions without considering a broader context, e.g., all the column names in the table & table description.

- Large Language Models (LLMs) are **powerful data generators** for table metadata, enabling the automatic creation of descriptive information about table & columns
- Leveraging all available metadata** is essential for generating better column descriptions as the quality of generated column descriptions influences the task performance
- The performance can be further improved by re-ranking as our current approach embeds these descriptions directly without **more fine-grained re-ranking** to refine the results